SECURITY REPORT 2026

Presented By
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Title: Vulnerability Assessment Report
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Cyber Security CIN ID: FIT/APR26/CS7650

EXECUTIVE SUMMARY

As part of my Future Interns Cyber Security internship, I conducted a passive vulnerability assessment on the public testing website httpbin.org. Using Nmap, OWASP ZAP in passive mode, and Wireshark, I identified several low-to-medium risk issues. The site is designed for educational purposes and is generally well configured, but a few improvements would strengthen its security posture in a production environment.

METHODOLOGY

- Nmap for comprehensive port scanning, service version detection, and OS fingerprinting.
- OWASP ZAP passive scanning to identify web application issues without active attacks.
- Wireshark for real-time capture and analysis of network traffic.

KEY FINDINGS

- **Open Ports 80 and 443 (Nmap**) – Risk: Low Both HTTP and HTTPS ports are listening. While acceptable for a test API, port 80 should automatically redirect to HTTPS in any live environment to prevent unencrypted traffic.

- **Missing Security Headers (ZAP Alerts**) – Risk: Medium Headers such as Strict-Transport-Security (HSTS) and X-Content-Type-Options were not present. This could allow downgrade attacks or MIME-type sniffing by malicious actors.

- **TLS 1.2 Still Supported (Wireshark)** – Risk: Medium The server negotiates both TLS 1.3 and the older TLS 1.2 protocol. Although most connections used 1.3, continued support for 1.2 slightly reduces the overall encryption strength.

- **Unencrypted HTTP Traffic (Wireshark)** – Risk: Low Plain HTTP requests and responses (including 200 OK and 404 pages) were captured in clear text, confirming that sensitive information should never be sent over port 80.

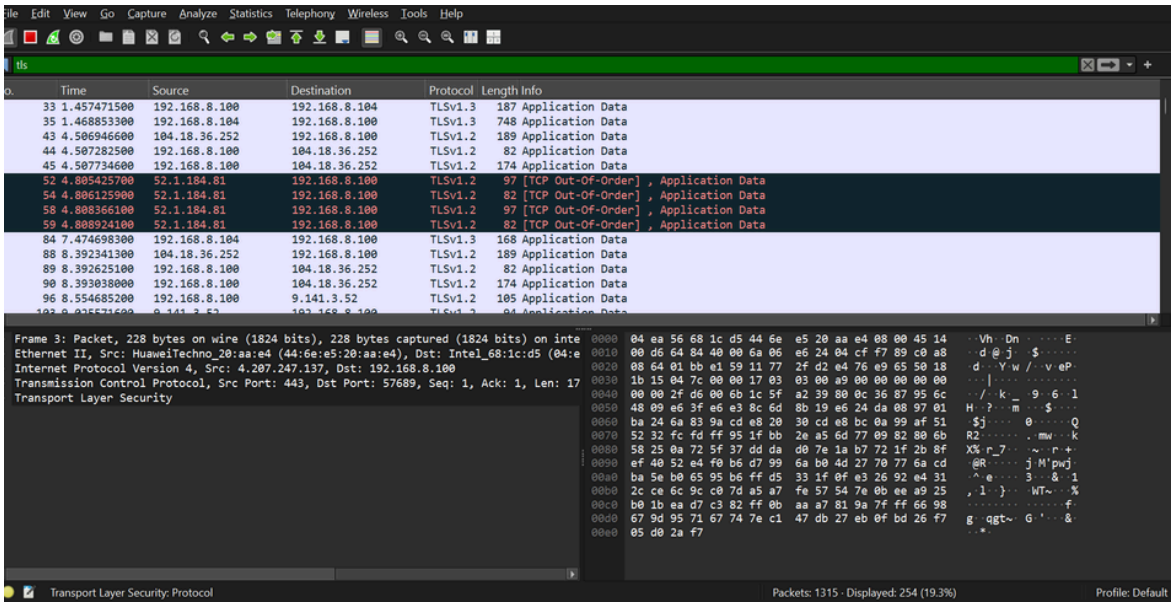
RISK CLARIFICATION TABLE

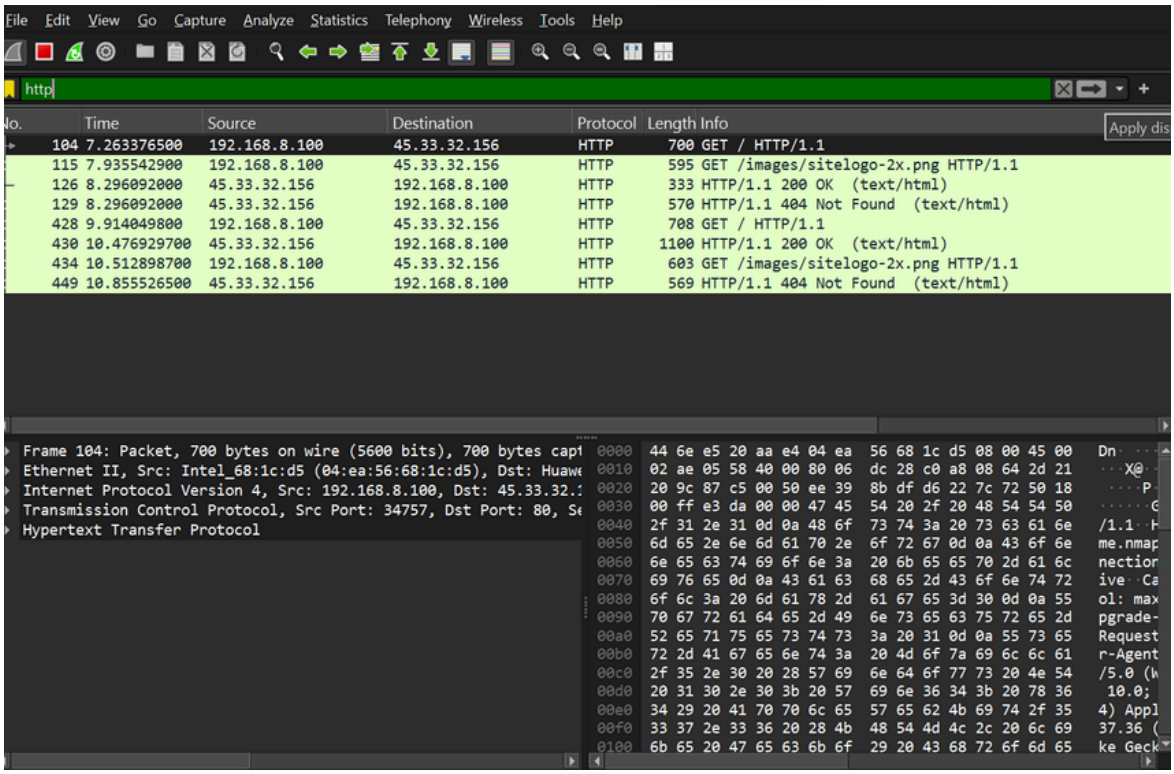
FINDING	RISK LEVEL	POTENTENTIAL BUSINESS IMPACT
Open Port 80	low	Minor risk of unencrypted traffic
Missing HSTS Header	medium	Possible downgrade attacks
TLS 1.2 Support	medium	Slightly weaker encryption in some cases
Plain HTTP Response	low	Data exposure if sensitive info is sent

Recommended Remediation Steps

- Implement an automatic HTTP-to-HTTPS redirect and enable HSTS with a long max-age value.
- Disable TLS 1.2 unless required by legacy systems.
- Add modern security headers (HSTS, X-Content-Type-Options, X-Frame-Options).
- For any production API, enforce encrypted traffic only.

SCREENSHOTS





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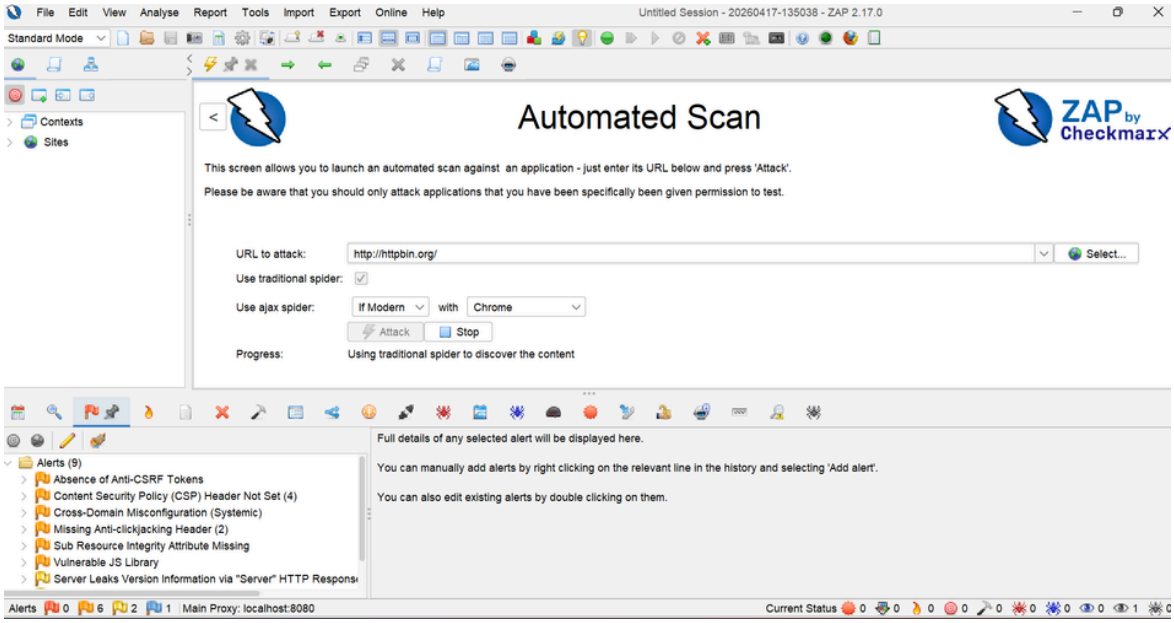

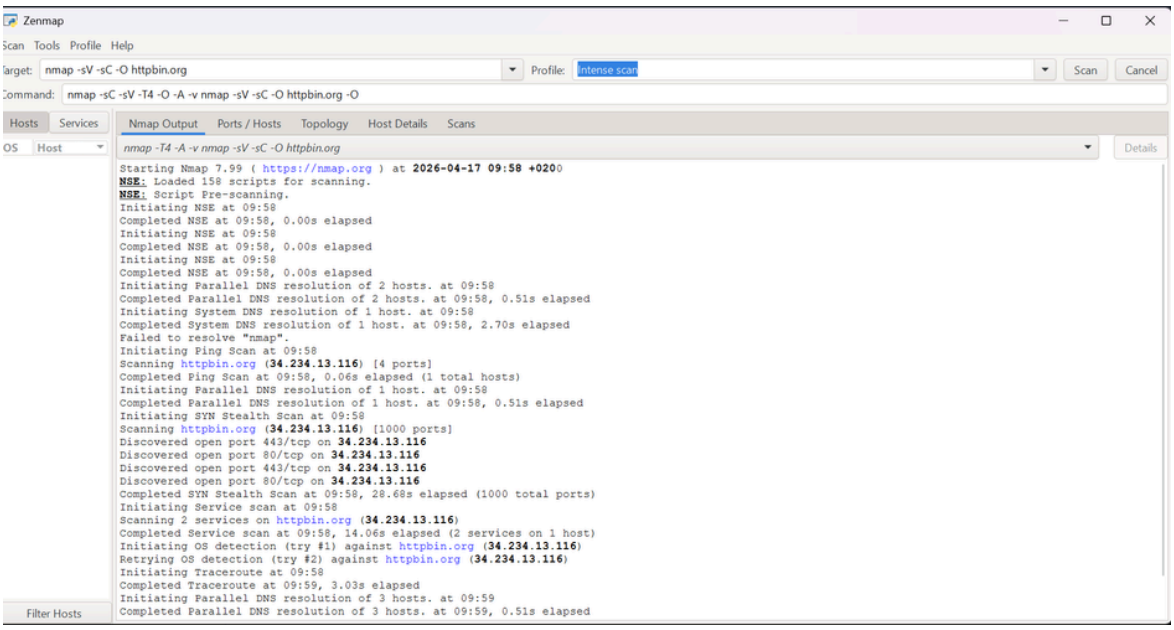

conclusion

This exercise gave me practical experience with real security tools and helped me understand how to present technical findings in clear business language. All identified issues are straightforward to fix.